

# Area of Regular Polygons

Lesson 5-9

**Name:**           Type your name

**Date:**           Today's date

**Class:**           Class period

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## TODAY'S OBJECTIVES

**Content Objective:** I can find the area of a regular polygon by decomposing it into triangles.

**Language Objective:** I can explain my steps using the words regular polygon, decompose, triangle, and composite.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



## WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Area of Regular Polygons

Regular Polygon

Decompose

Triangle

Composite

Formula



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

The space inside a regular polygon, found with  $A = (\frac{1}{2}bh) \times n$ , is the

2

A polygon with all sides and all angles equal is a .

3

To break a shape into smaller shapes is to  it.

4

A three-sided polygon is a .

5

Made of two or more shapes combined, a figure is .

6

A math rule written with symbols is a .

## Write About the Math

### How much will the gazebo flooring cost?

*¿Cuánto costará el piso del quiosco?*

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Area of Regular Polygons**

*Área de polígonos regulares*

☐ **Regular Polygon**

*Polígono regular*

☐ **Decompose**

*Descomponer*

☐ **Triangle**

*Triángulo*

☐ **Composite**

*Compuesto*

**Model:** We decompose the hexagon into triangles because we already know how to find a triangle's area. — Students explain that splitting the hexagon into 6 equal triangles lets them use the triangle area formula they already know, then combine the parts.

### Start

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **One triangle's area is \_\_\_\_ sq ft. The octagon has \_\_\_\_ triangles, so total area = \_\_\_\_ sq ft. The cost is \_\_\_\_ x \$3 = \$ \_\_\_\_.**
- **My answer is \_\_\_\_ because \_\_\_\_.**

*Mi respuesta es \_\_\_\_ porque \_\_\_\_.*

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### Build

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My evidence is \_\_\_\_, so \_\_\_\_.**  
*Mi evidencia es \_\_\_\_, entonces \_\_\_\_.*

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### Explain

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

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☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.